

1. Define a class `Car` with the following attributes: `brand`, `model`, and `year`. Include a method `display_info()` that prints the car's details. Create an object of this class and display the information.
2. Create a base class `Person` with attributes `name` and `age`. Derive a subclass `Student` that adds an attribute `student_id` and a method `display_student_info()`. Create an object of the `Student` class and display both person and student details.
3. Create two classes `Dog` and `Cat` with a method `sound()` that prints the sound made by each animal. Write a function that calls the `sound()` method regardless of the object type.
4. Create a class `Employee` with an instance variable `name` and a class variable `company_name`. Create multiple employee objects and demonstrate the difference between instance and class variables.
5. Write a function `check_even_or_odd()` that takes a number as input and checks whether the number is even or odd. Print the result.
6. Write a recursive function `factorial()` to calculate the factorial of a given number.
7. Write a function `sum_of_squares()` that takes a list of numbers and returns the sum of their squares using a for loop.
8. Write a function `print_fibonacci(n)` that prints the first `n` numbers of the Fibonacci sequence using a while loop.
9. Write a function `classify_age()` that takes an age as input and prints whether the person is a child, teenager, adult, or senior based on the age.
10. Write a function that takes a dictionary of student names and scores as input. Return a list of students who scored more than 70.
11. Create a list of numbers `[1, 2, 3, 4, 5]`. Use `map()` and a lambda function to return a list of their squares.

12. Create a list of numbers [10, 15, 20, 25, 30]. Use filter() and a lambda function to return only the numbers divisible by 5.
13. Use reduce() and a lambda function to calculate the product of the numbers in the list [1, 2, 3, 4].
14. Write a list comprehension to generate a list of the squares of all even numbers between 1 and 20.
15. Write a for loop that prints numbers from 1 to 50 that are divisible by 3.
16. Write a program using a while loop that prints the sum of all even numbers between 1 and 100.
17. Write a program to iterate through a given string and count how many vowels (a, e, i, o, u) it contains.
18. Write a program that uses a loop to print numbers from 1 to 10 but skips printing the number 5 and stops the loop when it reaches 8.
19. **Using split() Function:** Write a Python function split_sentence() that takes a sentence as input and splits it into a list of words.
20. **Using join() Function:** Write a Python function join_words() that takes a list of words as input and joins them into a single string with spaces between the words.